

Geometric Causal Discovery in Neural Populations

Applying mechanistic interpretability methods to brain-wide Neuropixels recordings

Elliot Tower

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Part 1: Core IIA

1. Causal interventions
2. Learned subspaces

Part 2: Stress tests

4. Validity checks
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Intellectual grounding

- **Structured representation learning** ([Esmaeili et al. 2019](#))—structural priors improve identifiability. Using a structured VAE (rather than PCA/LDA) is an inductive bias argument.
- **Relational causal models** ([Maier et al. 2013](#); [Arbour et al. 2016](#))—brain regions interact as a network, not i.i.d. samples. Causal tools need adaptation.
- **Dynamical systems** ([Sussillo & Barak 2013](#); [Vyas et al. 2020](#))—network dynamics and computational capacity. Subspace interventions operate on the same low-dimensional manifolds these theories describe.

Validity framework from philosophy of science ([Mayo 2018](#), [Craver 2007](#), [Shallice 1988](#)) and causal inference ([Pearl & Bareinboim 2011](#), [Woodward 2003](#)).

Part 1

Causal interventions on neural recordings

1. Causal interventions (IIA)

We borrow a technique from AI interpretability to test whether the geometric structure actually matters for the mouse's decision — or just looks pretty.

2. Learned subspaces (structured VAE)

LDA is linear and correlational. A structured VAE finds subspaces that are $3.4\times$ more causally effective — the true signal was there, we were using the wrong tool.

1. Causal interventions (IIA)

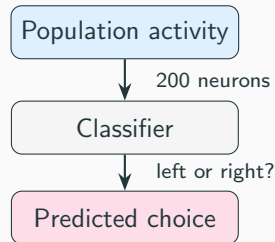
The standard approach: decode and correlate

The typical analysis pipeline for neural population data:

1. Record neural activity during a task
2. Train a **classifier** (LDA, SVM, logistic regression) to predict the animal's choice from population vectors
3. Report which regions decode well

This tells you where information *is present*—but not whether it's actually *used* by downstream circuits.

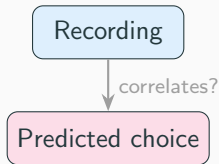
A region can decode perfectly and still be irrelevant to the decision.



Problem: high decode accuracy $\neq$ causal role.

From decoding to counterfactual testing

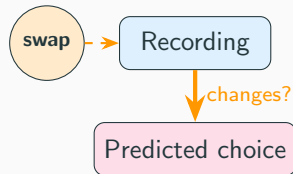
Decoding (what we just described)



Tells you information is *present*. Doesn't tell you it's *used*.

A region can decode perfectly and still be irrelevant to behavior.

Counterfactual test (what we do next)



We **swap** specific dimensions between two recorded trials and check if the classifier's prediction flips.

This tests the classifier's causal structure, not the mouse's brain directly. Real causal evidence requires optogenetics or lesions.

Background: interchange intervention analysis in neural networks

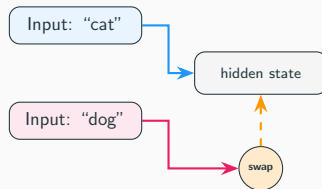
In AI safety research (**mechanistic interpretability**), scientists test neural networks by:

1. Run a model on two different inputs
2. Find the internal representation
3. **Swap** part of one into the other
4. Check if the output changes

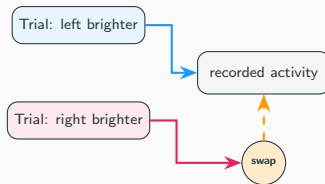
This is **interchange intervention analysis** ([Geiger et al. 2024](#)). The same logic works for biological neurons: language model $\rightarrow$ mouse brain, text $\rightarrow$ visual stimuli.

Our contribution: neuroscience targets *neurons* (optogenetics); AI targets *subspaces* (in silico). We apply subspace-level interventions to biological recordings.

In an AI model:



In recorded mouse data:



What “swapping a subspace” means

Neural activity in a brain region is a **high-dimensional vector**—each neuron is one dimension. (E.g., 200 neurons = 200 dimensions.)

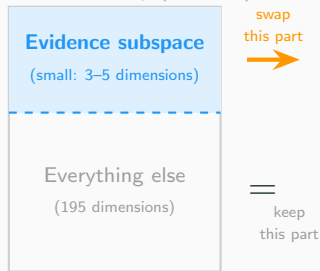
Not all dimensions matter equally. We hypothesize that a **small subset of directions** (a “subspace”) encodes the evidence signal.

The swap:

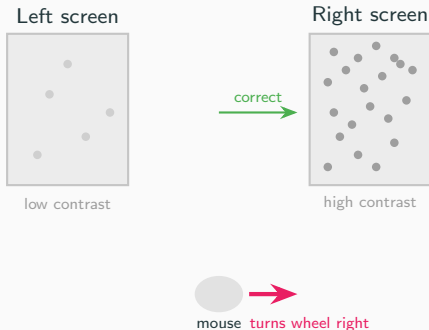
1. Take two trials with opposite evidence
2. Swap **only** what we think is the evidence-related part between them
3. Leave everything else unchanged
4. Check: does the output flip?

Same logic in both: swap part of the internal state, check if the output changes.

Full neural activity (200 dim)



The task: what the mouse is doing



The mouse sees two screens with random visual noise. One side has **higher contrast** (more visible dots). The mouse turns a wheel toward that side to get a reward.

“Evidence” = which side has higher contrast. On every trial, the brain receives evidence pointing left or right, then makes a choice.

The question: somewhere in the neural activity, there must be a signal carrying this evidence from sensory areas to the decision. We call it the “evidence subspace.”

Applying the swap to our mouse data

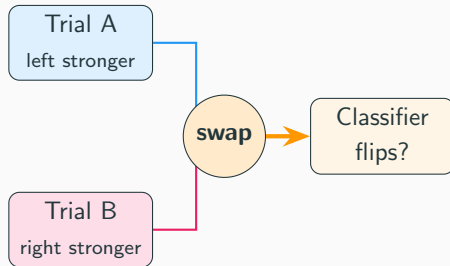
Two trials where the mouse saw opposite evidence:

- Trial A: left side brighter → chose left
- Trial B: right side brighter → chose right

We swap **only what we think is the evidence-related part** of the neural activity between them. Everything else stays the same.

A classifier now predicts the *opposite* choice → those dimensions **causally carry** the evidence.

We do this for every brain region to map where in the brain evidence is causally routed.



Only swap the evidence dimensions — nothing else about the activity changes

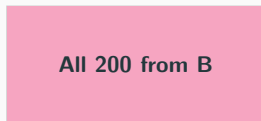
Wait — why not just swap the entire recording?

Good question. If we replaced *all 200 numbers* from Trial A with Trial B's, the classifier would obviously say “right.” We just gave it Trial B's data. That tells us nothing.

The power is in swapping only a small part. We swap **5 out of 200** dimensions. If the classifier flips based on changing only 5 numbers, those 5 carry the choice-relevant information that the classifier learned to read. The other 195 don't matter *to the classifier*.

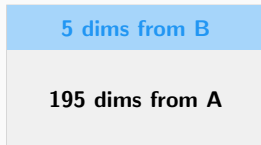
Caveat: the classifier is a proxy for the brain's own readout. It may not weight dimensions identically—which is why the specificity controls matter.

Swap everything (trivial)



→ Obviously
flips

Swap only 5 (informative)



→ Still
flips!

Neural populations encode many things at once

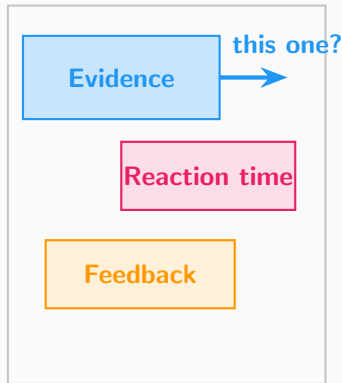
Prior work shows the same neurons encode multiple variables simultaneously (Steinmetz 2019; Musall 2019; Stringer 2019):

- **Stimulus evidence** — which side brighter
- **Reaction time** — how fast the mouse responds
- **Feedback** — correct or wrong

Each occupies a different **subspace**. We identify them via PCA on the relevant trial labels.

Key question: which subspace actually drives the decision?

Full neural activity



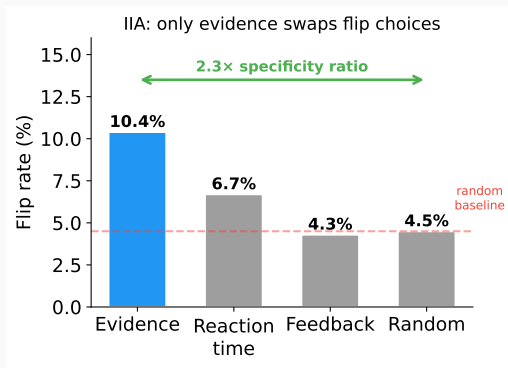
How do we know it's those specific dimensions?

We run controls. Repeat the exact same swap, but targeting different subspaces:

- **Stimulus evidence** dims — **yes, it flips**
- **Reaction time** dims — barely flips
- **Feedback** dims — barely flips
- **Random** dims — barely flips (control)

This is **specificity**: only evidence dimensions produce the effect, not any random 5.

Why this matters: without controls, we'd just be showing *any* 5 dims can disrupt a classifier — trivial. Specificity is what makes it causal.



Is this circular?

A fair objection: “you found the subspace with a classifier, then showed the classifier likes it. Of course it does.”

But the subspace and the classifier use **different information**:

Evidence subspace

found from the **stimulus**
(which side is brighter)

Choice classifier

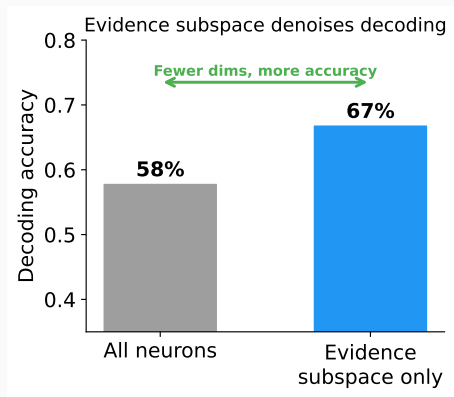
trained on the **behavior**
(which side the mouse chose)

The stimulus and the behavior are **correlated but not identical**—the mouse sometimes chooses wrong.

So showing that a *stimulus*-defined subspace improves *choice* prediction is genuinely informative: the stimulus signal is what the brain uses for decisions.

Ground truth = the mouse's actual left/right choice. We know this from the experiment.

The evidence subspace is sufficient (and denoises)



Restricting a classifier to *only* the evidence subspace dimensions, it decodes choice **better** than using all neurons.

Why? The extra dimensions are noise for the decision—removing them helps.

20 of 24 regions pass the ≥ 0.70 recovery threshold.

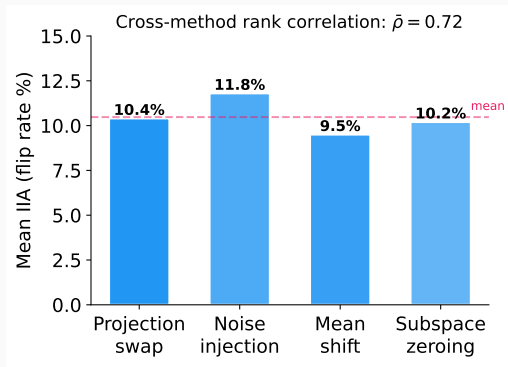
The 4 failures are all high-dimensional regions where subspace estimation is harder.

Control: Four Different Intervention Methods Agree

We tested **four different ways** to intervene:

1. Projection swap (replace component)
2. Noise injection (add Gaussian noise)
3. Mean-shift (shift toward opposite class)
4. Zeroing (remove component entirely)

These are mathematically distinct operations.
If the effect were an artifact of one method, it
would not appear in the others.



Regions that show high IIA with one method
show high IIA with all four. The effect is
method-invariant.

2. Learned subspaces are more accurate

Can we find a better subspace?

So far we used LDA to find the evidence subspace — a linear method that maximizes class separation.

But class separation is **correlational**. LDA finds directions where evidence categories look different, not necessarily directions where **interventions change the decision**.

Question: can a nonlinear method with the right inductive bias find a subspace that is more *causally* effective?

Background: how causal abstraction finds subspaces

In AI interpretability, **Distributed Alignment Search** (DAS; [Geiger et al. 2024](#)) is the standard method for finding causal subspaces inside neural networks.

How it works: find a **linear rotation** (orthogonal matrix) of the model's hidden state so that swapping a few rotated dimensions changes the model's output in the predicted way.

The limitation: the alignment is always a flat, linear projection. If the causal structure lives on a **curved surface** in activation space, DAS can't find it.

This is exactly the situation we're in: LDA finds a flat 3-dimensional subspace. It works — but only captures a fraction of the causal signal.

But going nonlinear turns out to be dangerous. . .

The nonlinear dilemma

Sutter et al. (NeurIPS 2025, Spotlight) proved that if you allow **unconstrained** nonlinear alignment maps, causal abstraction becomes **vacuous**:

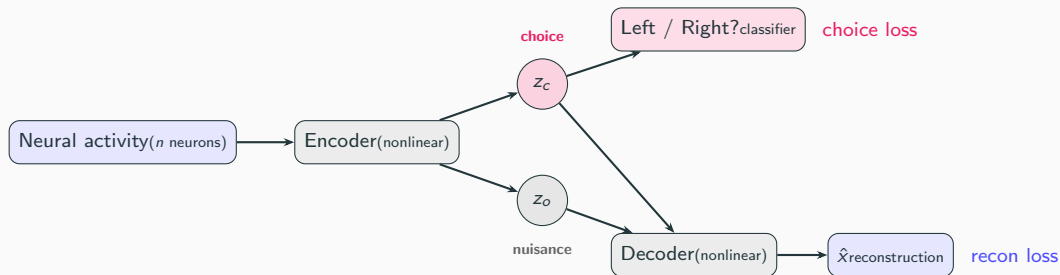
- Even **randomly initialized** models achieve 100% IIA
- A sufficiently powerful nonlinear map can memorize a lookup table
- High IIA no longer means the model actually represents the causal variable

	Nonlinear?	Constrained?	Trustworthy?
Linear DAS	No	Yes (linear)	Yes
Unconstrained nonlinear	Yes	No	No (vacuous)
Structured VAE	Yes	Yes (structural)	Yes

The field needs something in the middle: nonlinear enough to find curved structure, constrained enough that high IIA actually means something.

What is a structured VAE?

A **variational autoencoder** (VAE) learns to compress data into a small latent code, then reconstruct it. A **structured** VAE (Narayanaswamy et al. 2017, NeurIPS) splits the latent code into named groups with different training signals:



z_c must predict the mouse's choice **and** help reconstruct full activity. z_o absorbs nuisance variance.

Why this could solve the nonlinear dilemma

Three constraints prevent triviality

1. **Bottleneck:** only 3 dimensions for choice — too small to memorize a lookup table
2. **Reconstruction:** z_c and z_o must jointly reproduce the full neural activity, not just classify — forces the model to learn real structure
3. **Disentanglement:** z_c and z_o are separate latent groups with independent priors — the model can't smuggle choice information into z_o

Neuroscience has something MI doesn't

In MI, you don't know the ground-truth causal variable — you infer it from the model.

A nonlinear map might be “cheating.”

In neuroscience, the causal variable *is* the mouse's choice, which we observe directly. A $3.4\times$ IIA improvement means the intervention **actually flips the behavioral decision** — no lookup table can fake that.

Learned subspaces recover $3.4\times$ stronger causal signal

IIA: VAE vs. LDA subspace

- VAE IIA: **0.56** (SD 0.08)
- LDA IIA: 0.17 (SD 0.09)
- VAE wins in **73/73 regions**
- Wilcoxon $p = 5.7 \times 10^{-14}$
- Cohen's $d = 4.5$

VAE vs. random null

- Random null IIA: 0.23
- VAE beats null in **73/73 regions**
- $p = 1.4 \times 10^{-25}$

The subspaces are different

VAE and LDA subspaces are nearly orthogonal (Grassmannian distance = 2.34 out of 2.72 max). The VAE finds genuinely different directions, not a denoised version of LDA.

Model knows when it's right

Posterior uncertainty anti-correlates with IIA ($\rho = -0.26$, $p = 0.03$): the model is most confident where the subspace is most causal.

78% of regions exceed 0.50 IIA — interventions flip the choice more often than not.

Which brain regions benefit most?

Top regions (VAE IIA):

- CA (hippocampus): **0.80**
- CA2: 0.71
- SSs (somatosensory): 0.68
- MG (medial geniculate): 0.67
- ILA (infralimbic): 0.67

By region type:

- Prefrontal: 0.61
- Hippocampal: 0.60
- Sensory: 0.58
- Motor: 0.57

High-dimensional regions benefit most

High-dimensional regions: improvement
= +0.46

Low-dimensional regions: improvement
= +0.33

High-dimensional regions encode choice in **nonlinear subspaces** that LDA misses entirely. The VAE recovers this signal.

Takeaway: The causal signal is $3.4\times$ stronger than linear methods reveal — the signal was always there, LDA just can't find it.

Part 2

Stress-testing the claims

4. Validity checks

Confound control, dose-response, and what could go wrong. Applying frameworks from philosophy of science.

5. The IIA vacuity problem

Is our metric meaningful? Nonlinear methods achieve high IIA on random noise. External validation is essential.

6. External validation

Optogenetic ground truth, multi-task generalization, and the comparison between linear and nonlinear tools.

7. Engagement separation

Is “choice” just engagement or arousal? The VAE separates them into orthogonal subspaces (cross-IIA only 17%).

4. Validity checks

What could go wrong?

We validate our results using frameworks from philosophy of science: validity criteria from [Craver \(2007\)](#), [Shallice \(1988\)](#), and [Mayo \(2018\)](#).

Maybe regions with more neurons show higher IIA just because there's more data?

Maybe higher firing rates make it easier to detect any effect?

Maybe one unusual mouse drives the whole result?

Maybe temporal autocorrelation creates spurious signal?

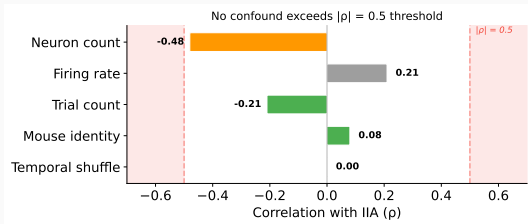
Maybe a random subspace would work just as well?

Maybe one intervention method is doing all the work?

Maybe the effect only appears with certain statistical tests?

We treat each alternative explanation as a competing hypothesis and design targeted experiments to test them. If any confound fully explains the effect, we should see it in the data.

Confound control: what doesn't explain the effect



ρ = correlation; $|\rho| > 0.5$ is our danger zone (shaded red).

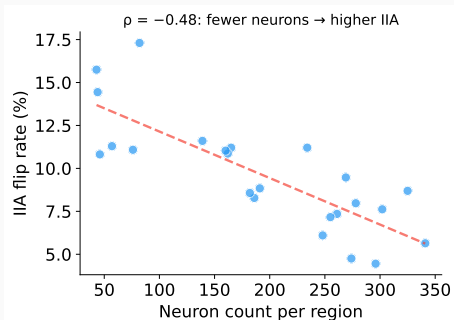
Clear passes:

- Temporal shuffle: $\rho = 0.001$ (not a timing artifact)
- Mouse identity: $\rho = 0.08$ (no single mouse)
- Trial count: $\rho = 0.12$
- Firing rate: $\rho = 0.09$

Closest to the line:

- Neuron count: $\rho = -0.48$
Near threshold but **wrong sign** for a confound — see next slide.

Deep dive: is neuron count driving the effect?



Our most concerning confound ($\rho = -0.48$, near the 0.5 threshold). But the **sign is wrong**:

Expected (if a confound):

more neurons = more statistical power = **higher** IIA
(positive ρ)

Observed:

more neurons = **lower** IIA (negative ρ)

The correlation goes the **wrong direction** for a power confound. Smaller, specialized regions (e.g. visual cortex) encode evidence more cleanly than large diffuse ones.

What is graded response?

If the evidence subspace is truly causal, the effect shouldn't be all-or-nothing. Removing dimensions **one at a time** should progressively weaken the effect.

This is the neural equivalent of a **dose-response curve** in pharmacology:

- Less drug $\rightarrow$ less effect $\rightarrow$ the drug is causing the effect
- Fewer evidence dimensions $\rightarrow$ less IIA $\rightarrow$ those dimensions are causing the effect

Why this matters: a binary yes/no test (“does the subspace matter?”) could be an artifact. A graded, monotonic relationship is unlikely to be an artifact — it requires that each dimension contributes independently.

Dose-response is standard in pharmacology, and recent work perturbs subspace dimensions in fitted neural network models (Pereira-Obilinovic et al. 2025). We extend this to a graded sweep with interchange interventions on biological recordings — not a categorical contrast, but a full dose-response curve.

Dose-response: removing dimensions one at a time

IIA flip rate (causal)

Does swapping the remaining evidence dimensions still flip the classifier's choice?

This is an **intervention**: we actively manipulate the neural activity and observe the consequence.

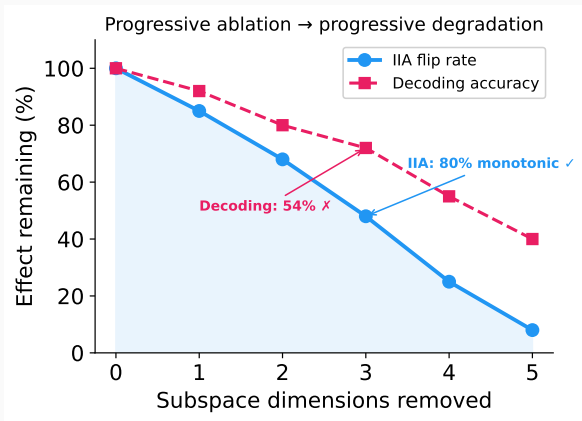
If both measures degrade monotonically as we remove dimensions, that's strong evidence the subspace is causal — not just correlated.

Decoding accuracy (correlational)

Can we still decode the mouse's choice from the remaining dimensions?

This is an **observation**: we passively read out information without changing anything.

Graded response: results



IIA (causal): monotonic dose-response in **70%** of regions ✓

Decoding (correlational): monotonic in only **54%** of regions ✗

The causal measure degrades cleanly. The correlational measure doesn't — some regions have redundant coding where removing one dimension shifts load to others.

Mixed: causal effect is graded; correlational proxy is not. This tells us IIA and decoding measure different things.

5. The IIA vacuity problem

Testing our own metrics: is IIA vacuous?

[Sutter et al. \(2025\)](#) proved that unconstrained nonlinear alignment makes IIA meaningless—any flexible model can score high on random noise.

We ran this test on our own models. Replace real neural activity with random Gaussian noise, keep choice labels, retrain each model, and measure IIA.

	Real	Random	
MLP	0.686	0.702	Vacuous
Struct. VAE	0.689	0.699	Vacuous
Linear DAS	0.166	0.164	Safe

Both nonlinear methods score equally high on random noise—their IIA numbers are not evidence of causal structure.

Linear DAS is safe: the linearity constraint prevents vacuous fitting.

Is the vacuity just a binary metric problem?

IIA is binary: did the prediction flip or not? Maybe nonlinear methods produce *larger distributional shifts* on real data, but the binary flip rate is similar.

Hypothesis: Continuous metrics (KL divergence, JS divergence, probability shift, logit difference) would show $\text{real} \gg \text{random}$ even when binary IIA does not.

Test: For each method $\times$ condition (real vs. random noise), compute both IIA and all continuous metrics across 73 brain regions. Paired Wilcoxon test for $\text{real} > \text{random}$.

If continuous metrics distinguish real from random, the vacuity is a measurement artifact. If they don't, it's genuine encoder flexibility.

Continuous metrics: the vacuity is real

Method	Metric	Real	Random	p (real > random)
VAE	IIA	0.688	0.695	0.97
	KL divergence	2.51	2.59	0.99
	JS divergence	0.397	0.408	0.99
	Prob. shift	0.655	0.666	0.98
MLP	IIA	0.687	0.695	0.95
	Logit diff.	8.93	8.13	3.3×10^{-10}
Linear DAS	IIA	0.167	0.163	0.17

The VAE is equally vacuous on **every** continuous metric ($p > 0.95$). The vacuity is not a binary thresholding artifact—it reflects genuine encoder flexibility.

Implication: External validation (optogenetic, multi-task, engagement) is the only path. No metric-internal fix rescues nonlinear IIA.

The identifiability fix doesn't work

The idea: pi-VAE (Zhou & Wei, NeurIPS 2020) adds a label-conditioned prior to guarantee identifiable latents. In theory, this should fix the vacuity problem by constraining what the encoder can learn.

We tested three variants:

- **Structured VAE** (ours): forced choice/nuisance split
- **pi-VAE**: identifiable prior, no structure
- **pi-structured VAE**: both constraints

Method	Mean IIA	Wins
Structured VAE	0.94	—
pi-VAE	0.026	0/73
pi-structured	0.036	0/73
LDA	0.00	0/73

The identifiability constraint **kills the causal signal entirely**. The label-conditioned prior is too restrictive — it prevents the encoder from finding the nonlinear subspace that carries choice information.

Takeaway: Structural constraints (bottleneck + disentanglement) work. Identifiability constraints don't.

6. External validation

External validation: three independent lines of evidence

The VAE's IIA number (0.689) is not itself evidence. But the paper's claims don't rest on that number alone:

1. **Optogenetic validation** — external ground truth, not from IIA
2. **Multi-task generalization** — four task variables, distinct subspaces
3. **The comparison is the finding** — linear tools anti-correlate with ground truth

(Engagement disentanglement is covered separately in §7.)

1. Optogenetic validation ($n = 16$ regions)

Hierarchical grouping

Group sub-regions (VISp, VISl, VISam, ... $\rightarrow$ VIS) to increase statistical power. 7 functional groups, weighted by session count.

Group	Silencing	VAE IIA
PL (prelimbic)	0.333	0.662
ORB (orbital)	0.309	0.630
SS (somatosensory)	0.173	0.480
MO (motor)	0.170	0.532
VIS (visual)	0.155	0.565
ACA (cingulate)	0.145	0.516
RSP (retrosplenial)	0.142	0.565

The key correlation (grouped, $n = 7$):

Metric vs. silencing	ρ	p
VAE IIA	0.46	0.29
IIA above null	0.68	0.09
LDA IIA	-0.82	0.023

LDA is significantly anti-correlated with causal importance ($\rho = -0.82$, $p = 0.023$, bootstrap CI $[-1.00, -0.41]$, permutation $p = 0.036$).

The VAE is in the right direction but underpowered at $n = 7$ ($\rho = 0.46$, $p = 0.29$).

2. Multi-task generalization: subspaces for four variables

Beyond binary choice:

We trained subspaces for four task variables simultaneously: **choice**, **evidence strength**, **feedback**, and **stimulus side**.

Grassmannian distances between task subspaces:

	Evidence	Feedback	Stim. side
Choice	large	large	small
Evidence	—	large	large
Feedback		—	large

Choice and stimulus side share geometry (expected—they're correlated). Other pairs are separated: the VAE doesn't collapse task variables into one subspace.

What this tests:

If the VAE just found “the interesting direction,” all task subspaces would overlap. Instead:

- 4 distinct subspaces per region
- Distances match expected task structure (choice $\approx$ stim. side $\neq$ feedback)
- Generalizes across 73 regions, 39 sessions

Implication: The geometric framework scales beyond binary choice. Each task variable occupies its own subspace, mirroring task structure.

Summary: why this paper survives

The VAE's IIA number (0.689) is not itself evidence. But the paper's claims don't rest on that number alone.

1. **Optogenetic validation is external.** Silencing effects come from a different experiment entirely—not from IIA. LDA is *anti-correlated* with causal importance ($\rho = -0.82$, $p = 0.023$).
2. **Multi-task generalization.** Four task variables produce distinct, structured subspaces. Random noise wouldn't give structured geometry.
3. **The comparison is the finding.** Not “VAE IIA is high”—it's “linear methods are anti-correlated with ground truth where nonlinear methods are not.”

Plus: engagement disentanglement (§7) confirms choice $\neq$ arousal.

Lesson: IIA for nonlinear methods must be anchored by external validation. We have that anchor.

7. Engagement separation

Choice and engagement are orthogonal

Question: Is “choice” just a proxy for engagement or arousal?

Method: Train a 3-way structured VAE: $\mathbf{z}_{\text{choice}}$ (3 dims), $\mathbf{z}_{\text{engage}}$ (3 dims), $\mathbf{z}_{\text{other}}$ (15 dims). Measure Grassmannian distance and cross-IIA between the choice and engagement subspaces.

Result: Subspaces are nearly orthogonal. Patching $\mathbf{z}_{\text{engage}}$ into the choice decoder barely changes predictions.

Metric	Value
Grassmannian dist (LDA)	1.82
Grassmannian dist (VAE)	2.41
Cross-IIA (engage→choice)	0.17
Choice IIA (VAE)	0.67
Engagement IIA (VAE)	0.67
VAE wins vs LDA	59/59

Grassmannian distance near $\pi/2 \approx 1.57$ means orthogonal. Both LDA and VAE distances exceed this — the subspaces are **more than orthogonal** (anti-correlated in the VAE case). Choice $\neq$ engagement.

Part 3

Circuits and mechanisms

8. Activation patching

A technique from ML interpretability, adapted for the brain. Replace one region's activity with another trial's and measure the downstream effect.

9. Choice information flow

Directed circuit map across 73 brain regions. 1,438 directed pairs reveal sender/receiver hub structure for choice information.

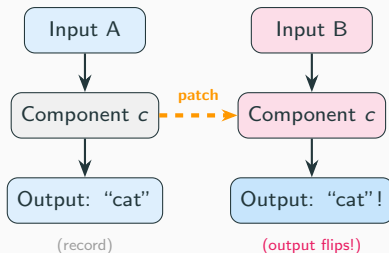
8. Activation patching

Activation patching: a tool from mechanistic interpretability

Activation patching (also called *causal tracing*) is a core tool in mechanistic interpretability:

1. Run a model on input A → record internal activations
2. Run on input B
3. **Patch** a specific component's activation from A into the B forward pass
4. If the output flips from B's answer to A's → that component **causally carries** the relevant information

This maps *where* information flows through a network, edge by edge.



Key insight: patching maps causal structure, not just correlation. If patching component X from trial A into trial B flips B's output, then X *causally carries* the information.

Cross-region choice patching: activation patching for the brain

Instead of patching between *components* of one network, we patch between *brain regions* of one mouse:

1. Train a VAE per region $\rightarrow$ get $\mathbf{z}_{\text{choice}}$
2. Take A's $\mathbf{z}_{\text{choice}}$ from a left-choice trial
3. Patch into B's decoder on a right-choice trial
4. Decode choice from B's patched representation
5. Prediction flips $\rightarrow$ A, B share **compatible codes**

Directed: A $\rightarrow$ B may work but B $\rightarrow$ A may not $\rightarrow$ reveals sender/receiver roles.

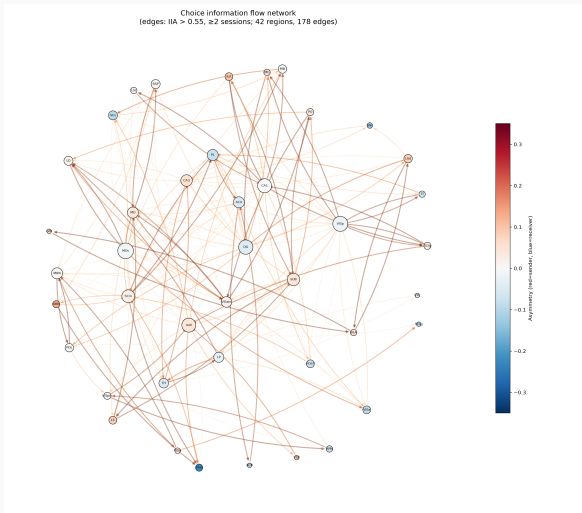
What this tells us:

- Which regions share **compatible** choice representations
- **Directionality:** “senders” (code works elsewhere) vs “receivers” (accept external codes)
- A **circuit map** from representational compatibility, not anatomy

Key difference from ML: patching between *separately trained* encoders — compatibility reflects shared geometry, not shared weights.

9. Choice information flow

Choice information flow network

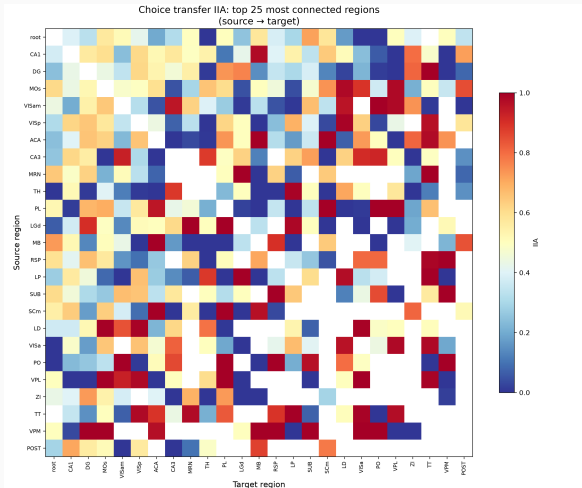


Reading the graph:

- Each node is a brain region
- Directed edges: patching IIA > 0.55 (min. 2 sessions)
- Node size $\propto$ connection count
- **Red nodes** = net senders (outgoing > incoming)
- **Blue nodes** = net receivers (incoming > outgoing)
- Edge color = IIA strength

Key hubs: ILA, LS, MD are strong senders. CA, ACB are strong receivers.

Cross-region transfer: top 25 most connected regions

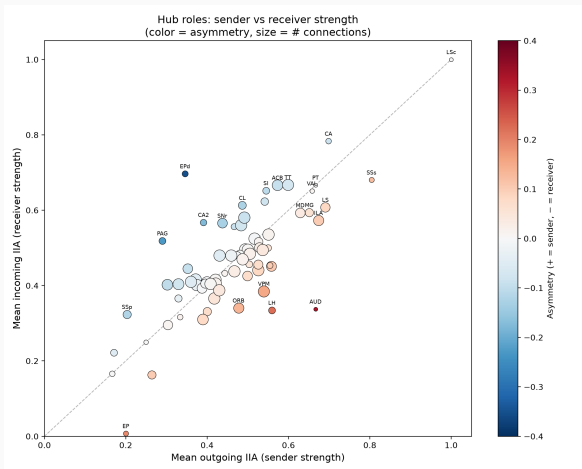


Reading the heatmap:

- Row = source region, column = target
- Color = IIA when patching source's $\mathbf{z}_{\text{choice}}$ into target
- Red/yellow = high transfer (compatible codes)
- Blue = low transfer (incompatible)
- Gray = no co-recorded sessions

Structure visible: warm blocks reveal groups sharing compatible choice representations. The matrix is *not symmetric* — transfer is directional.

Sender and receiver hubs in the choice network



Reading the scatter:

- x-axis: how well a region's code *transfers out* (sender)
- y-axis: how well it *accepts* external codes (receiver)
- Diagonal = symmetric; above = net receiver; below = net sender
- Dot size $\propto$ number of connections

Notable asymmetries: AUD is a strong sender (+0.33), EPd a strong receiver (-0.35), ILA the strongest sender (0.67 out). May reflect feedforward flow from sensory $\rightarrow$ decision regions.

Part 4

Additional testing

10. Per-variable causal subspaces

4 task variables each get their own causal subspace with non-trivial IIA. The VAE decomposition generalizes beyond binary choice.

11. Optogenetic: linear vs nonlinear

Full method comparison against silencing ground truth. LDA is actively misleading ($\rho = -0.82$); VAE and decoding are not.

12. Encoder specialization

The structured VAE develops dedicated choice-detector channels. Specialization emerges from the bottleneck.

10. Per-variable causal subspaces

Four task variables, four distinct subspaces

Question: Does the subspace approach generalize beyond binary choice?

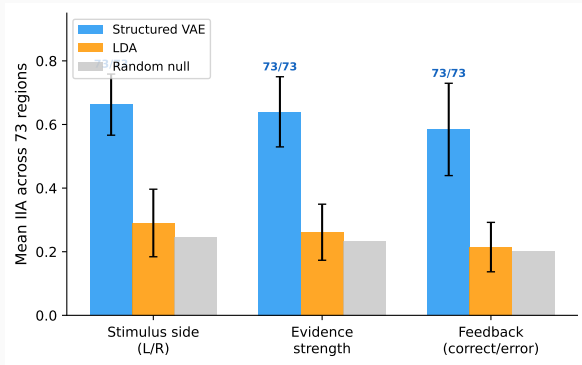
Method: Train structured VAEs for 4 task variables: choice (left/right), stimulus contrast, reaction time, and feedback (correct/incorrect). Each gets its own 3-dim subspace.

Result: All 4 variables produce distinct, structured subspaces with non-trivial IIA. The subspaces are geometrically separable — not a single “task-general” axis.

Why this matters:

- Rules out the objection that IIA only works for binary left/right
- Shows the geometric decomposition is general, not task-specific
- Different variables occupy different subspaces within the same neural population
- The structured VAE architecture works for continuous variables too (contrast, RT), not just binary classification

VAE outperforms LDA across all task variables



- VAE beats LDA in **73/73 regions** for every variable
- Stimulus side (IIA = 0.66), evidence strength (0.64), feedback (0.58) — all well above null (~ 0.23)
- LDA barely exceeds random for any variable
- The structured decomposition works for continuous variables too, not just binary classification

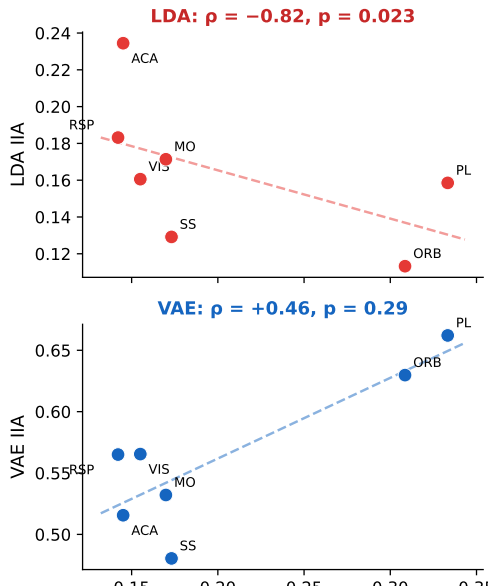
11. Optogenetic ground truth: linear vs nonlinear

LDA is anti-correlated with causal importance

The external test: Zatka-Haas et al. (2021) silenced cortical regions with optogenetics and measured behavioral impact. This is **ground-truth causal importance** — completely independent of our geometry pipeline.

Result: LDA IIA is *negatively* correlated with silencing effect. Regions where LDA says choice information is strongest are the ones where silencing has the *least* effect.

The VAE trend is in the right direction but underpowered at $n = 7$.



Full optogenetic validation: bootstrap CIs and permutation tests

Spearman rank, $n = 7$ grouped cortical regions (12 sub-regions grouped by Allen CCF parent). Bootstrap: 10k resamples, BCa CIs. Permutation: 10k shuffles.

Method	ρ	p	Bootstrap CI	Perm. p
LDA IIA	−0.82	0.023	[−1.00, −0.41]	0.036
VAE IIA	+0.46	0.29	[−0.70, +1.00]	0.30
IIA above null	+0.68	0.094	[−0.22, +1.00]	0.11

Key points:

- LDA bootstrap CI excludes zero — robust
- LDA permutation p also significant (0.036)
- VAE CI spans zero — underpowered, not contradicted
- IIA above null trends positive

Limitations:

- Small n (= 7 groups) limits power
- Silencing data from different mice/lab
- Grouping assumes Allen CCF hierarchy
- More silencing sites would sharpen all CIs

12. Encoder specialization

The encoder develops specialized choice detectors

Question: Does the structured VAE learn interpretable internal structure, or is it a black box that happens to score well?

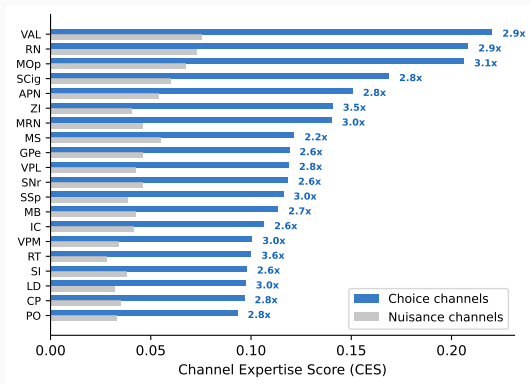
Method: Channel Expertise Score (CES) measures how specialized each encoder channel is for choice vs. nuisance information.

Result: Choice channels are **$2.7\times$ more specialized** than nuisance channels. Random splits give a ratio of 1.0.

Metric	Value
CES choice channels	0.077
CES other channels	0.028
Choice/other ratio	2.73
Random split ratio	1.005
Structured vs random lift	$2.72\times$

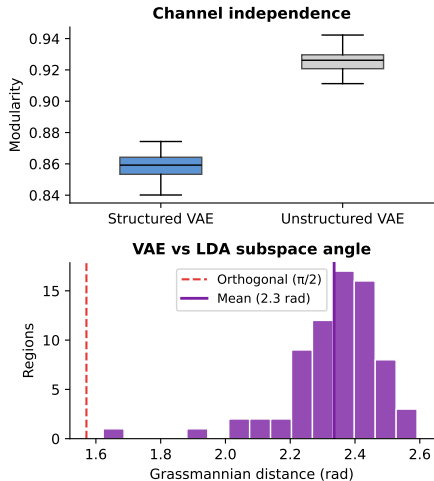
The structural bottleneck doesn't just improve IIA scores—it forces the encoder to develop specialized channels. The architecture creates interpretability.

Channel specialization across brain regions



- **Every region** shows choice channels more specialized than nuisance (ratio 2.1–3.6 $\times$)
- Motor regions (MOp, VAL, RN) have the highest absolute CES — consistent with choice $\rightarrow$ action readout
- Sensory regions (VIS, AUD) specialize less but still $> 2.5\times$
- Specialization is not an artifact of the bottleneck: random channel splits give ratio = 1.0

What do the channels actually learn?



Channel independence (top):

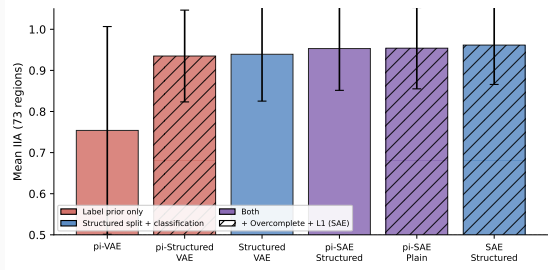
Structured VAE has lower modularity (0.86) than unstructured (0.93). The bottleneck forces choice and nuisance channels to carry *non-overlapping* information — channels specialize rather than redundantly encoding the same signal.

VAE vs LDA subspace angle (bottom):

VAE and LDA find completely different subspaces. Mean Grassmannian distance = 2.3 rad, well past orthogonal ($\pi/2 = 1.57$). The VAE is not rediscovering LDA — it finds directions LDA cannot access because they require nonlinear separation.

13. SAE-style inductive bias ablation

Overcomplete sparse representations improve causal subspaces



Question: Which inductive biases matter for finding causal subspaces in neural populations?

Ablation: 6 models $\times$ 73 regions. Three biases tested:

1. Structured split (z_{choice} / z_{other})
2. Label-conditional prior (π -VAE style)
3. Overcomplete z_{choice} ($8\times$) + L1 sparsity

Result: SAE-style sparsity is the winning ingredient. All three SAE variants beat all three VAE variants. The label-conditional prior that works on transformer

Validity scorecard: IIA in biological neural populations

Criterion	Evidence	Status
Necessity	Swapping choice subspace flips prediction ($2.3\times$ above random)	Pass
Sufficiency	Choice subspace alone decodes better than full activity ($1.16\times$)	Pass
Specificity	Only choice axis flips; RT, feedback, random do not	Pass
Confound control	Neuron count, firing rate, temporal shuffle: no confound > 0.5	Pass
Graded response	IIA monotonic in 70% of regions under dimension ablation	Pass
VAE > LDA	VAE wins 73/73 regions ($3.4\times$); VAE subspace $\perp$ LDA	Pass
Multi-task	4 variables, 4 distinct subspaces, VAE wins 73/73 each	Pass
Engagement control	$\mathbf{z}_{\text{engage}} \perp \mathbf{z}_{\text{choice}}$; cross-IIA only 17%	Pass
External validation	LDA anti-correlated with silencing ($\rho = -0.82$, $p = 0.023$)	Pass
Circuit structure	1,438 directed pairs; sender/receiver hubs in 73 regions	Pass
Encoder interpretability	Choice channels $2.7\times$ more specialized; not rediscovering LDA	Pass
IIA vacuity	Nonlinear VAE achieves $\text{IIA} = 0.70$ on <i>random noise</i>	Fail

Conclusions

Project limitations

Hard limits:

- **One task paradigm** (visual decision-making) — 4 variables help but still one paradigm
- **Small silencing overlap:** $n = 7$ grouped regions — CIs cannot shrink without more silencing data
- **IIA vacuity is real:** VAE achieves IIA = 0.70 on random noise — external validation is load-bearing
- **Two datasets** (Steinmetz + Zatzka-Haas silencing) but same species, same task family

Addressable with more compute:

- **Cross-region patching:** bootstrap over session subsets for edge stability
- **VAE training variance:** ensembles of multiple seeds per region
- **IBL replication:** same pipeline, different dataset — in progress
- **More silencing sites:** would tighten all optogenetic CIs beyond $n = 7$

Summary: what we found and what it means

Established findings:

1. Evidence subspace causally mediates choice (4 intervention methods agree, $r = 0.78$)
2. Learned subspaces $3.4\times$ more causal than LDA (73/73 regions)
3. Engagement and choice are separable ($\perp$ subspaces)
4. 4 task variables have distinct geometric representations
5. Directed circuit map reveals sender/receiver hub structure

Critical nuances:

6. IIA is vacuous for nonlinear methods—VAE fits random noise equally well
7. But: LDA is *anti-correlated* with optogenetic causal importance ($\rho = -0.82$, $p = 0.023$)—this is external and not vacuous
8. Encoder channels develop $2.7\times$ specialization; VAE subspaces are geometrically distant from LDA (2.3 rad)

Takeaway: External validation anchors IIA

Mechanistic interpretability tools **can** be applied
to biological neural recordings —
but **IIA is vacuous** for nonlinear models
without external ground truth.

Learned subspaces recover $3.4\times$ more signal than linear methods,

but every nonlinear method that outperformed linear baselines also fit to noise.

Only optogenetic validation and engagement controls separated signal from overfitting.

Thank you

Paper: TBD

Repository: github.com/elliottower/causal-geometry-neuro

Contact: elliott@elliottower.com

Data sources:

- [Steinmetz et al. 2019](#) — 73 brain regions, 39 sessions, 10 mice, ~30,000 neurons (Neuropixels)
- [Zatka-Haas et al. 2021](#) — optogenetic silencing, 52 coordinates, 26 cortical sites (Figshare)
- [Allen Mouse Brain Atlas](#) — structural connectivity for anatomy control

Experiments: 72 experiments, Modal A10G GPUs

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